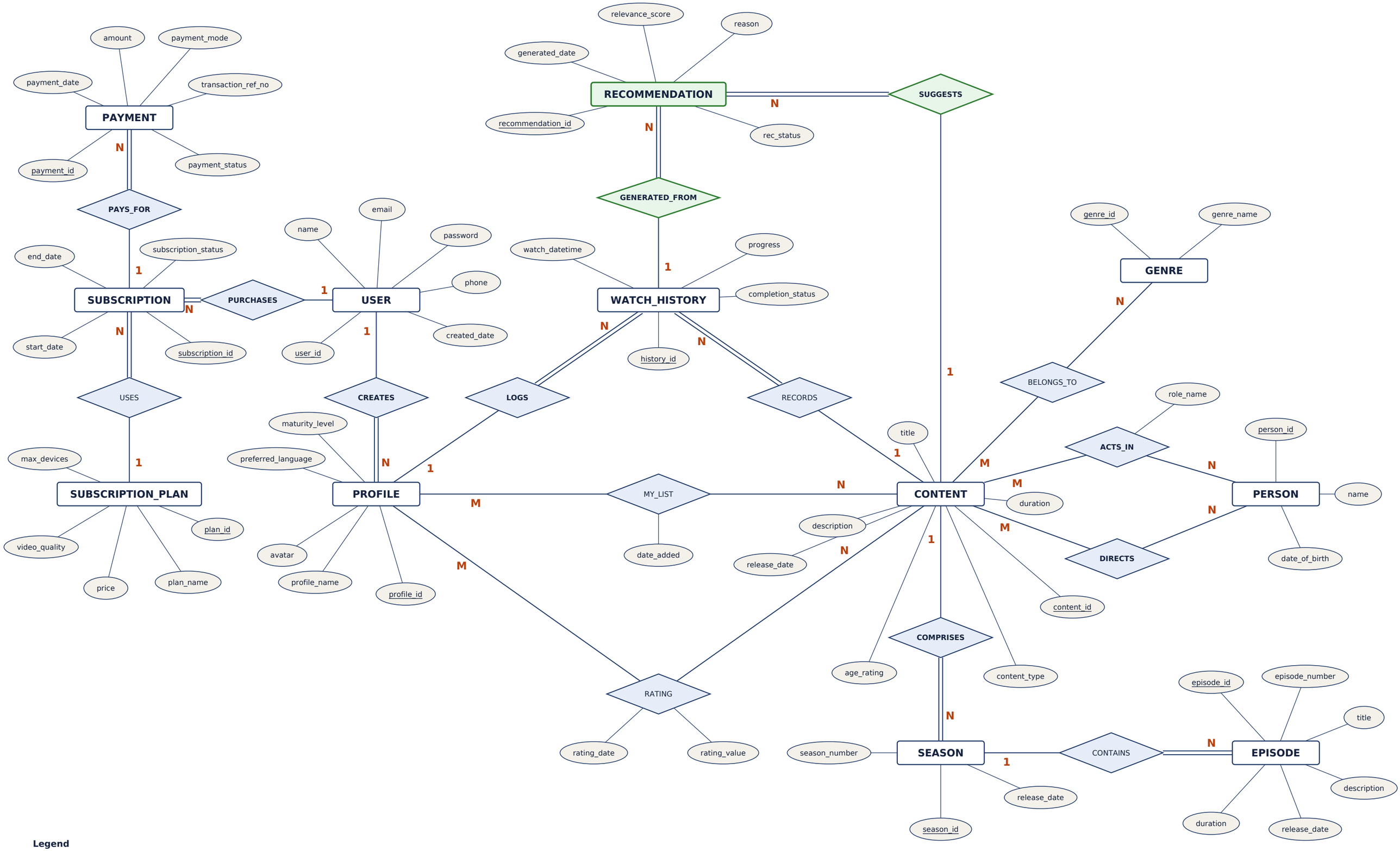
Streaming Management System — ER Diagram (Revised)



Legend

Rectangle = entity ◇ Diamond = relationship Ellipse = attribute underlined = key attribute
Double line = total participation Bold relationship name = renamed from HAS / HAS_*
Green = new: RECOMMENDATION entity generated from WATCH_HISTORY and suggesting CONTENT

Relational Schema (mapped from the revised ER diagram)

Strong entity -> table with its key as PK. 1:N relationship -> FK on the N side. M:N relationship -> separate table whose PK is both FKs (MY_LIST, RATING, CONTENT_GENRE, CONTENT_CAST, CONTENT_DIRECTOR).

ACCOUNT & BILLING

USER

user_id	name	email	password	phone	created_date
		UNIQUE			

PROFILE

profile_id	user_id	profile_name	avatar	preferred_language	maturity_level
	FK → USER				

SUBSCRIPTION_PLAN

plan_id	plan_name	price	video_quality	max_devices

SUBSCRIPTION

subscription_id	user_id	plan_id	start_date	end_date	subscription_status
	FK → USER	FK → SUBSCRIPTION_PLAN			

PAYMENT

payment_id	subscription_id	payment_date	amount	payment_mode	transaction_ref_no	payment_status
	FK → SUBSCRIPTION				UNIQUE	

CONTENT CATALOGUE

CONTENT

content_id	title	description	release_date	content_type	duration	age_rating

GENRE

genre_id	genre_name
	UNIQUE

CONTENT_GENRE

content_id	**genre_id**
FK → CONTENT	FK → GENRE

SEASON

season_id	content_id	season_number	release_date
	FK → CONTENT		

UNIQUE (content_id, season_number)

EPISODE

episode_id	season_id	episode_number	title	description	release_date	duration
	FK → SEASON					

UNIQUE (season_id, episode_number)

PERSON

person_id	name	date_of_birth

CONTENT_CAST

content_id	**person_id**	role_name
FK → CONTENT	FK → PERSON	

CONTENT_DIRECTOR

content_id	**person_id**
FK → CONTENT	FK → PERSON

VIEWING ACTIVITY

WATCH_HISTORY

history_id	profile_id	content_id	watch_datetime	progress	completion_status
	FK → PROFILE	FK → CONTENT			

MY_LIST

profile_id	**content_id**	date_added
FK → PROFILE	FK → CONTENT	

RATING

profile_id	**content_id**	rating_value	rating_date
FK → PROFILE	FK → CONTENT		

RECOMMENDATION (new)

recommendation_id	history_id	content_id	generated_date	relevance_score	reason	rec_status
	FK → WATCH_HISTORY	FK → CONTENT				

UNIQUE (history_id, content_id)

Legend: bold underlined = primary key (blue cell) orange cell with "FK → TABLE" = foreign key PK+FK cells = composite key made of foreign keys

RECOMMENDATION reaches PROFILE through history_id -> WATCH_HISTORY.profile_id, so profile_id is not repeated (avoids a transitive dependency, keeps 3NF).